Data Science and Big Data Analytics for Transportation Systems

Main Traininr: Surafel Luleseged (PhD)
Assistant: Abebe Lata (MSc)

AASTU

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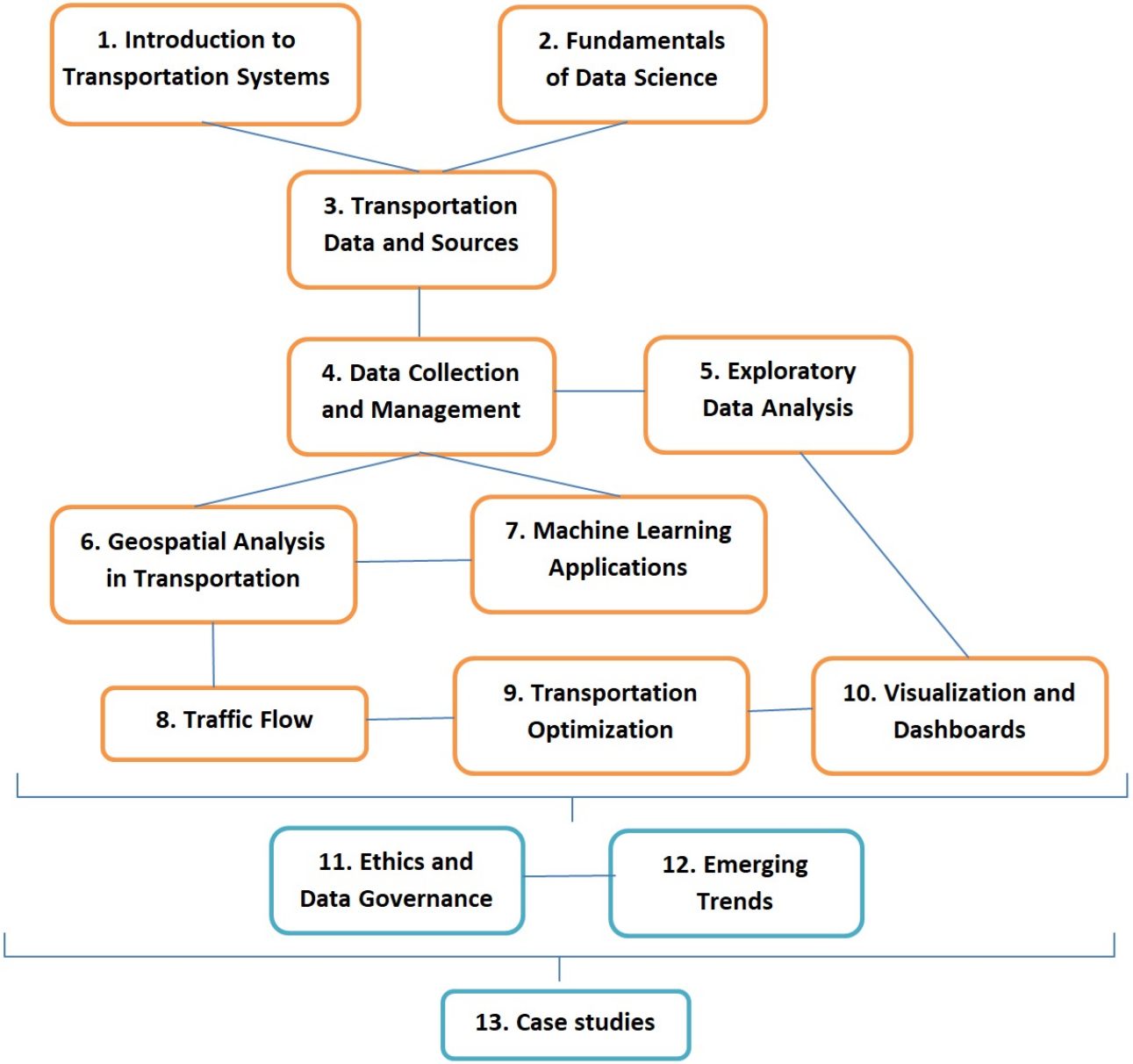
Content


1. Introduction to Transportation Systems
2. Fundamentals of Data Science
3. Transportation Data and Sources
4. Data Collection and Management
5. Exploratory Data Analysis
6. Geospatial Analysis in Transportation
7. Machine Learning Applications
8. Traffic Flow
9. Transportation Optimization
10. Visualization and Dashboards
11. Ethics and Data Governance
12. Emerging Trends
13. Case studies

Training details

- Objective
  - Understand key transportation data sources and data systems
  - Interpret and analyze transport data for decision-making
  - Apply data-driven approaches to traffic management and public transport planning
  - Use predictive analytics for congestion, demand, and safety assessment
  - Identify accident hotspots and support road safety interventions
  - Develop dashboards and communicate insights effectively using tools like Power BI
  - Support smart mobility and Intelligent Transportation Systems initiatives
  - Apply principles of data governance, ethics, and data sharing

Training details

- Mode of delivery
  - In person for 15 experts from Transportation office
  - A 3 days training (compact)
  - Active participation and group work is expected
  - Traineers are expected to come with their own laptop for the traininig
  - Software: Python, Excel, PowerBI
- Successful completion
  - A uccessful completion for getting the certificate needs an active participation and over 83% attendance
- Materials
  - Training module will be shared after refinement at the end of the training
  - Data will be shared for the simulation workouts
  - Any additional reference books can be refereed
- Contacts: Any question or additional information can be requested by directly contacting the trainer via surafel.luleseged@aastu.edu.et